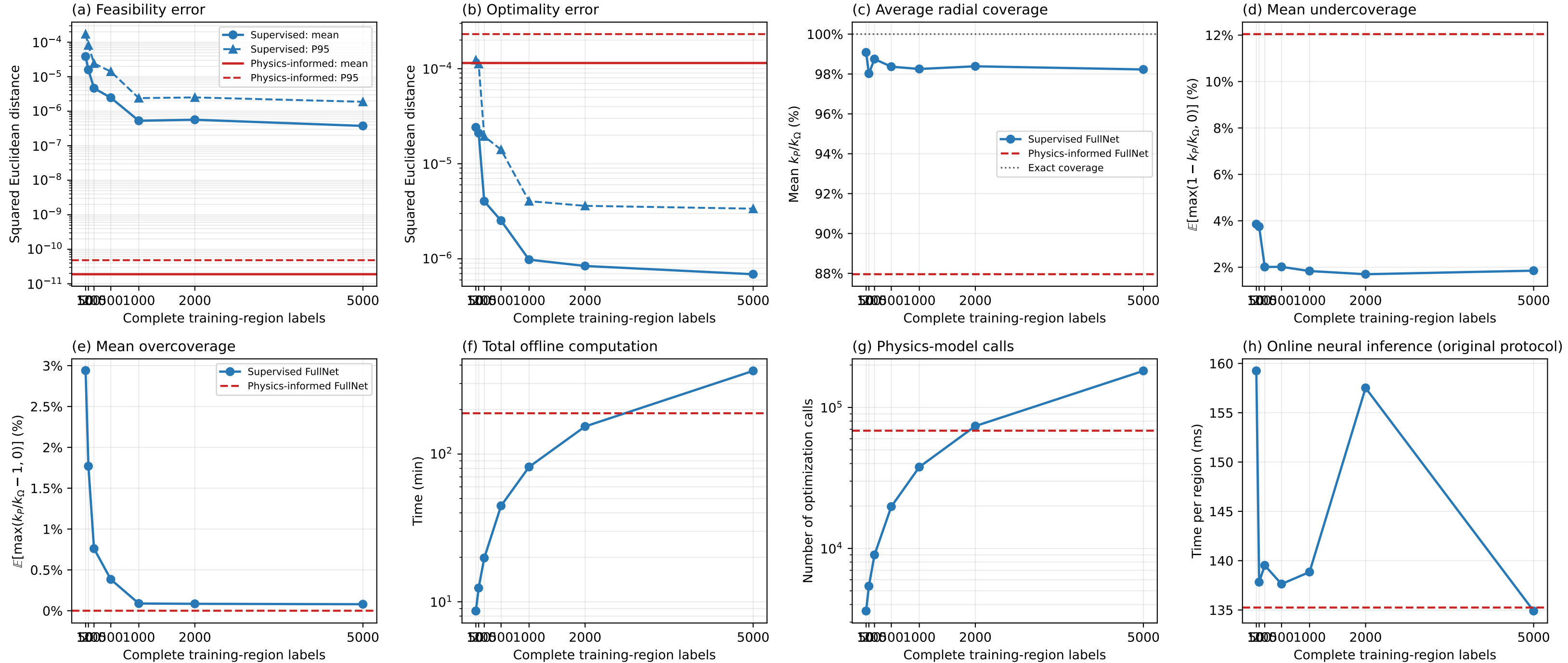


Same-architecture supervised baseline versus physics-informed model

Fixed-normal support-point labels



Errors are squared Euclidean distances; all methods use the same 50 test conditions and evaluation directions. Online time uses one end-to-end forward pass as in the original manuscript.